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OKAYA POWER LTD.

(SMF / VRLA BATTERY DIVISION)

A-42, Sector-63, Noida (UP) INDIA

Tel.: 0120-4640800 Fax: 0120-4640827

Email: vrla@okayapower.com

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Innovation expressway

- This car PC has a two-megapixel camera and in-built features of WiFi, GPS
- The motherboard has been shrunk to Q7 CPU module, making the PC very compact in size
- The car PC consumes one-fifth the power by a laptop
- Expected to be low-cost compared to the available DIN-based car PCs

is usually an issue. Besides, storage needs space and adds to the weight. But the car PC offers a 1 GB DDR2 on-board memory and 16GB of solid-state drive (SSD) storage, expandable to 64 GB. The use of data storage device that uses solid memory helped us reduce the weight and size of the gadget, making it a perfect fit for the car headrest," says Immanuel.

Though the company has not revealed the cost of the car PC, they said the price would be competitive compared to the standard DIN-based car PCs available in the market. iWave System Technologies plans to market the product through partnership with original equipment manufacturers (OEMs), automotive customers and independent software vendors (ISVs). "Leading cab companies are evaluating the car PC to offer free entertainment during long commutes and Internet access as value-added services," says Saliya.

Development challenges

It was six man-years of effort by iWave to develop this innovative device. And, it wasn't a smooth ride. "We faced numerous challenges like BOM optimisation, thermal management and finding partners for printed circuit board (PCB) fabrication of the Q7 CPU module PCB that involves the complicated HDI technology not yet available in India," says Saliya.

While BOM optimisation was essential to maintain the price and performance balance of the gadget having over 100 major components, thermal management was crucial to

prevent the gadget failure.

Elaborating on the thermal management hurdle, Immanuel says, "A car PC needs a proper heat management, otherwise functionality of the device gets affected. The temperature inside a car is already high. Besides, the excess heat generated by the headrest made up of leather is another problem. To achieve thermal management, we went in for special thermal conducting coated heat spread materials."

The initial drive

But how did this innovation process begin for iWave? The company began focusing on design services in May 1999. Their first customer project was the design and development of a complex reference board based on 32-bit RISC processor. Initially, they faced hurdles like any other young and inexperienced team. However, two-three years of hands-on training proved a boon for iWave. Not only the young engineers mastered the technology, but also fine-tuned the focus of the company to meet the industry needs. About 30 per cent of their 131-strong engineering resources, which amounts to 200 man-years, were channeled towards R&D from 2004 onwards.

The road ahead

Looking ahead, iWave System Technologies is presently working on a low-power, high-performance mobile computing platform.

"A team of 10 engineers had been working on it for a year and it is production-ready now," says Saliya. The company believes that ultra-slim gadgets will be the future innovative technologies. "Ultra-slim gadgets that enable mobility, connectivity and provide more comfort in terms of infotainment and productivity to the consumers are the future electronic products," says Saliya, who believes the industrial, consumer, health care and automotive sectors to be the most receptive of next-generation computing solutions. ●